

Stable Causal Targeting with Machine Learning: Application to a Behavioral Field Experiment in Consumer Finance

Abstract

Many randomized field experiments exhibit heterogeneous but noisy treatment responses, making flexible machine-learning estimates of conditional average treatment effects poorly calibrated and treatment-prioritization rules unstable. We develop a practical subgroup-targeting workflow that extends Stable Discovery of Interpretable Subgroups via Calibration to a consumer-banking experiment. The procedure ranks candidate CATE estimators by validation-fold calibration slope, constructs nested equal-weight ensembles, and selects the ensemble with the largest out-of-fold treatment-effect contrast between its predicted top-10% subgroup and the remainder. In two simulation designs with known CATE, the selected ensemble ranks near the best individual estimator in ground-truth targeting performance. Applied to a large overdraft messaging experiment at a Turkish bank, the selected ensemble identifies a top-10% subgroup with a positive holdout treatment-effect estimate even though the sample average effect is negative. The other two treatment-arm comparisons do not produce positive holdout effects under the same procedure. These findings illustrate how calibration-based ensemble selection can support more stable personalized intervention decisions in low-signal behavioral settings.

Keywords: behavioral interventions, consumer banking, heterogeneity, model calibration, subgroup discovery

JEL codes: C52, C53

1 Introduction

Financial institutions routinely conduct randomized field experiments to evaluate the effectiveness of promotional and informational messages (e.g. Bertrand et al., 2010; Karlan et al., 2016; Alan et al., 2018). While these experiments are typically designed to estimate average treatment effects, a growing body of research documents heterogeneity in individual responses. For example, Bertrand et al. (2010) show that males respond more to creative content. Athey et al. (2025) show that reminders encouraging students to renew their financial-aid applications before a non-binding deadline are more effective for some individuals than for others. These studies illustrate that the same informational or promotional intervention can generate different responses across individuals, underscoring the importance of accounting for treatment effect heterogeneity.

Recent advances in causal machine learning have made it possible to estimate conditional average treatment effects (CATEs) and to use these estimates to guide individualized treatment assignment. Athey and Imbens (2016) introduce causal trees, which modify regression tree methods to optimize for goodness of fit in treatment effects and to account for honest estimation. Wager and Athey (2018) develop causal forests method, extending random forests to estimate heterogeneous treatment effects. Künzel et al. (2019) propose several meta-learner frameworks that decompose CATE estimation into regression sub-problems, which can be solved using any supervised learning method, called base learners. The S-learner estimates the outcome by using all of the features and the treatment indicator, without giving the treatment indicator a special role. The T-learner uses base learners to estimate the conditional expectations of the outcomes separately for treated and control

units, and then takes the difference. The X-learner uses the observed outcomes to estimate the unobserved individual treatment effects (ITEs), and then estimates the CATE function as if the ITEs were observed. Nie and Wager (2021) proposes another meta-learner method, the R-learner, which residualizes both the outcome and the treatment with respect to covariates, and then regresses the residualized outcome on the residualized treatment.

In customer retention contexts, Ascarza (2018) demonstrate that targeting based on predicted churn performs substantially worse than targeting based on estimated treatment effects, a finding echoed in subsequent marketing studies (Fernández-Loría and Provost, 2022). More recently, Athey et al. (2025) show that sending reminders to students with low predicted probability of renewing financial aid in the absence of the treatment can be counterproductive, and propose hybrid approaches that incorporate the strengths of predictive approaches (accurate estimation) and causal approaches (correct targeting criterion).

Besides heterogeneity, treatment effects in realistic samples are often noisy, resulting in globally poorly calibrated CATE estimates and potentially unreliable treatment allocation decisions. This is especially concerning in behavioral field experiments, where effects are typically modest and signal-to-noise ratios are low. These challenges are compounded when researchers search over many possible subgroups or model specifications to try to find subgroups with desirable average treatment effects. As emphasized in the clinical trial literature on data-driven subgroup discovery (see, for example, the reviews in Ondra et al., 2016; Huber et al., 2019), aggressive search without appropriate adjustment can easily overfit and yield subgroups that do not replicate.

To address this challenge, we extend the Stable Discovery of Interpretable Subgroups

via Calibration (StaDISC) approach (Dwivedi et al., 2020), originally developed for discovery of interpretable subgroups in clinical trials, and adapt it to causal targeting in behavioral field experiments in consumer finance. Our approach is grounded in the predictability, computability, and stability (PCS) framework (Yu and Kumbier, 2018, 2020). *Predictability* demands that models demonstrate out-of-sample prediction accuracy before their conclusions can be trusted. *Computability* requires that analyses be computationally feasible and reproducible. Importantly, *stability* insists that conclusions be robust to reasonable perturbations in data processing, model specification, and random sampling.

Specifically, we first conduct a predictive reality check via calibration, assessing whether a set of CATE estimation models can predict treatment effect heterogeneity on validation data. We then use a calibration-slope statistic to order the candidate estimators, construct a sequence of nested equal-weight ensembles from that order, and choose the ensemble that maximizes an out-of-fold top-10% targeting contrast. We report the selected ensemble’s holdout performance, while treating it as exploratory because selection-statistic development also consulted holdout diagnostics, as disclosed in Supplementary Section S1.

We apply our approach to a field experiment conducted by Alan et al. (2018) at Yapi Kredi (YK), one of the largest banks in Turkey. The experiment aimed to learn about the optimal strategy for advertising its overdraft product. YK randomly sent different promotional text messages to 108,000 marginal overdrafters who had not overdrafted during the previous few months. The outcome of interest was the usage of the overdraft product. We focus on comparing a bundling message that combines an overdraft discount with automatic bill payment against the baseline message that only states overdraft availability.

Although the average treatment effect is negative, the top-10% subgroup identified by the selected mean ensemble `{x_rf, r_rflasso}` yields a holdout treatment effect of 0.0617 (SE = 0.0346; $t = 1.78$), nominally significant at the 10% level.

In the two Monte Carlo designs, the selected ensemble ranks second among the 23 individual estimators plus the ensemble when methods are ordered by mean ground-truth top-10% holdout CATE. An intentional set of 40 second-stage scoring functions was examined, and calibration slope was compared with $\bar{B}_{0.9}$ as an alternative first-stage ordering score. Supplementary Section S1 documents their generation, evaluation, and the choice of the primary specification.

The rest of the paper is organized as follows. Section 2 describes the dataset from the YK study. Section 3 presents the CATE estimation models that we consider. Section 4 introduces a calibration-based approach to assess the accuracy and stability of these models, and presents calibration results. Section 5 defines the nested calibration-slope ensemble procedure and reports the final holdout evaluation. Section 6 concludes. Supplementary Section S1, placed after the references, documents scoring-function generation, simulation evaluation, and first-stage sensitivity; Supplementary Section S2 reports the other two treatment-arm comparisons.

2 Dataset from the YK Study

2.1 Motivation for Finding Heterogeneous Treatment Effects and Sample Selection

The YK experiment enrolled 108,000 checking account holders who were marginal overdrafters and had not overdrafted during the previous few months. Alan et al. (2018) found that, compared to sending no message, a text message simply stating overdraft availability (“We remind you that, for your immediate cash needs, you have a Flexible Account at Yapi Kredi with [fill]TL limit. Have a nice day.” (1TL = \$0.56 during the sample period)) *increased* average overdraft usage. In contrast, a message offering a 50% discount on the 60% annual percentage rate (APR) overdraft rate (“Use your Yapi Kredi Flexible Account and we will give you back half of the interest rate that is accrued between now and [date] as WP.” (WP = wallet points)) *decreased* average usage. This result supports the hypothesis that unshrouding overdraft costs reduces demand, as consumers tend to underestimate them and have limited attention; explicitly mentioning the price reminds them of the cost, thereby motivating efforts to avoid overdrafts.

Moreover, Alan et al. (2018) found that messages bundling the overdraft discount with automatic bill payment (“Authorize automatic bill payments from your account before [date], receive up to a maximum of 30TL WP. Use your Yapi Kredi Flexible Account and we will give you back half of the interest rate that is accrued between now and [date] as WP.”) or a discount on debit card (“Use your Yapi Kredi debit card and get back 5% of your purchases up to a maximum of 25TL WP between now and [date]. Use your Yapi

Kredi Flexible Account and we will give you back half of the interest rate that is accrued between now and [date] as WP.”) further decreased average usage. This enhanced effect likely occurs because mentioning overdrafts together with transactions that could trigger them provides a more powerful reminder to avoid such transactions.

However, the “shrouded attributes” framework of Gabaix and Laibson (2006) predicts that unshrouding interventions may yield heterogeneous responses, with at least two sub-groups potentially responding positively:

1. **Financially sophisticated consumers** who were already aware of overdraft availability. For these individuals, the reminder conveys no new information and may even decrease usage by making the cost of overdraft salient. However, some may increase usage if the reminder reduces the cognitive cost of accessing a funding source they had been overlooking.
2. **Liquidity-constrained consumers** with acute short-term funding needs. For this group, the reminder that overdraft is available may *increase* their overdraft usage by making salient a funding source they might not have actively considered, thereby lowering the cognitive barrier to borrowing.

Our main analysis compares the bundling message that combines an overdraft discount with automatic bill payment against the baseline message that only states overdraft availability. The full sample consists of 36,024 customers: 17,981 in the treatment group that received the bundling message and 18,043 in the control group that received the baseline message. The outcome variable is a binary indicator for overdraft usage, equal to 1 if the account balance was negative during the post-intervention period and 0 otherwise. The

difference-in-means estimate of the average treatment effect—that is, the mean outcome difference between treated and control units—is -0.0138 (standard error = 0.0049) for the full sample. This suggests that, relative to the baseline message, the bundling message led to a modest average decrease in overdraft usage. Given the potential heterogeneity in treatment effects, we aim to identify a subgroup of customers for whom the bundling message increases average overdraft usage compared to the baseline message.

To evaluate model performance, we randomly split the full sample by treatment status, allocating 20% (7,205 customers) to a holdout set. The holdout observations were excluded from estimator fitting and fold-level ensemble selection. During subsequent method development, however, holdout diagnostics were consulted across candidate statistics and treatment-arm comparisons. The final holdout result is therefore exploratory rather than an independent confirmatory test. Data on the remaining 28,819 customers are used for model training and validation.

2.2 Feature Engineering and Selection

The original dataset contains a rich set of features. These include binary indicators (e.g., indicators for randomization strata, reminder frequency, campaign type, and prior service activations) and continuous variables (e.g., assets, deposits, debt levels, minimum balance, and prior credit card usage). Since some variables contain missing values, we create dummy indicators to flag missingness and impute the missing values with zero. This results in a total of 519 features.

Given the high dimensionality of features, we use Lasso for feature selection to reduce the

risk of overfitting. We fit separate Lasso models within the treatment and control groups among 28,819 customers, standardizing the features so that coefficient magnitudes are comparable. The top 10 features with maximum absolute coefficients from each model are shown in Figure 1. Taking the union of these features yields a set of 14 selected features. We note that this Lasso-based feature selection step is an extension to the StaDISC workflow in Dwivedi et al. (2020), as the original methodology discretizes continuous features into binary ones and does not prescribe a specific feature selection procedure.

Table 1 presents an overview of the outcome variable and the features in the control and treatment groups in the full sample with 36,024 customers. Overall, the covariates are balanced across the two groups.

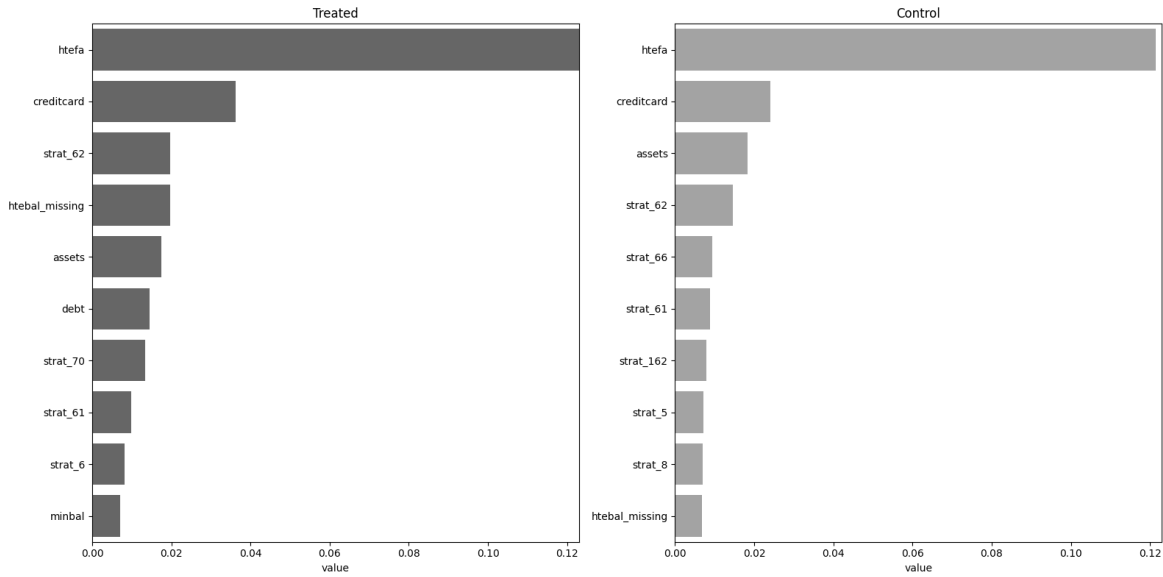


Figure 1: Absolute Lasso coefficients for feature selection. The left panel shows the top 10 features for the treated group; the right panel shows the corresponding top 10 features for the control group. The meaning of these features is presented in Table 1.

Table 1: Summary statistics for outcome and features by treatment and control groups (full sample).

Variable (abbreviation)	Control No. (%)	Treatment No. (%)
Outcome		
overdraft usage (fausebal=1)	5719 (31.7)	5451 (30.3)
Randomization Strata		
Male, married, low OD, other city, unknown bal (strat_5=1)	6 (0.0)	12 (0.1)
Male, married, low OD, other city, N/A bal (strat_6=1)	160 (0.9)	157 (0.9)
Male, married, low OD, other city, below-median bal (strat_8=1)	427 (2.4)	427 (2.4)
Male, unmarried, low OD, unknown city, N/A bal (strat_61=1)	252 (1.4)	251 (1.4)
Male, unmarried, low OD, unknown city, zero bal (strat_62=1)	787 (4.4)	788 (4.4)
Male, unmarried, low OD, other city, zero bal (strat_66=1)	326 (1.8)	324 (1.8)
Male, unmarried, low OD, Istanbul, zero bal (strat_70=1)	102 (0.6)	104 (0.6)
Female, married, low OD, unknown city, below-median bal (strat_162=1)	314 (1.7)	311 (1.7)
Prior Usage		
Used overdraft in 6 months pre-treatment (htefa=1)	3170 (17.6)	3094 (17.2)
Baseline account balance missing (htebal_missing=1)	119 (0.7)	110 (0.6)
	Control Mean (SD)	Treatment Mean (SD)
Financial Characteristics		
Avg monthly assets, Sept 2011–Aug 2012 (assets)	537.1 (1562.6)	563.9 (1805.8)
Avg monthly debt, Sept 2011–Aug 2012 (debt)	252.5 (670.3)	251.7 (650.5)
Minimum deposit balance pre-treatment (minbal)	93.9 (434.5)	92.6 (441.5)
Pre-randomization credit card usage (creditcard)	0.5 (0.5)	0.5 (0.5)

Note: For each binary variable, the count and the percentage of ones in each group are reported. For each continuous variable, mean and standard deviation in each group are reported. Low OD = overdraft limit < 1/2 minimum wage. bal = account balance.

2.3 Data Splitting for Model Training and Validation

Following Dwivedi et al. (2020), we do not create a separate validation set. Instead, we perform a 4-fold cross-validation (CV) by randomly splitting the data on 28,819 customers stratified by treatment status. We denote this split as cv_{orig} , which is used to tune the hyperparameters for all CATE estimation models via cross-validation.

A key innovation of PCS is its emphasis on stability across data perturbations. We generate two more stratified 4-fold CV splits, denoted by cv_0 , cv_1 . The hyperparameters tuned on cv_{orig} are applied when fitting models on the training folds of cv_0 and cv_1 , and the resulting models are applied to the validation folds of cv_0 and cv_1 for evaluation.

Note that for each 4-fold CV split, there are 4 distinct training-validation fold pairs. This results in a total of 12 training-validation fold pairs across cv_{orig} , cv_0 and cv_1 .

3 CATE Estimation Models

For unit i , let $Z_i \in \{0, 1\}$ denote the treatment indicator, $\mathbf{X}_i$ the feature vector, $Y_i(1)$ and $Y_i(0)$ the potential outcomes under treatment and control, and $Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$ the observed outcome. The average treatment effect (ATE) is defined as

$$ATE := E[Y(1) - Y(0)].$$

The individual treatment effect (ITE) for unit i is defined as

$$D_i := Y_i(1) - Y_i(0),$$

which is unobservable. The responses under control and treatment are defined as

$$\mu_0(\mathbf{x}) := E[Y(0)|\mathbf{X} = \mathbf{x}], \quad \mu_1(\mathbf{x}) := E[Y(1)|\mathbf{X} = \mathbf{x}].$$

The conditional average treatment effect (CATE) is defined as

$$\tau(\mathbf{x}) := E[D \mid \mathbf{X} = \mathbf{x}] = E[Y(1) - Y(0) \mid \mathbf{X} = \mathbf{x}] = \mu_1(\mathbf{x}) - \mu_0(\mathbf{x}).$$

Following Dwivedi et al. (2020), we train CATE estimation models under four meta-learner frameworks: S-learners, T-learners, X-learners, and R-learners. Their base learners are drawn from Lasso, penalized logistic regression, random forest (RF), and gradient-boosted trees (denoted by XGB because the XGBoost implementation is used). We also consider causal trees and causal forests. The full set of estimation models is as follows.

1. *S-learners* (2 models). A S-learner (Künzel et al., 2019) estimates the response function $\mu(\mathbf{x}, z) := E[Y \mid \mathbf{X} = \mathbf{x}, Z = z]$ using all units. Let $\hat{\mu}(\mathbf{x}, z)$ denote the resulting estimator. The CATE for an individual with features $\mathbf{x}_i$ is then estimated as $\hat{\tau}(\mathbf{x}_i) = \hat{\mu}(\mathbf{x}_i, 1) - \hat{\mu}(\mathbf{x}_i, 0)$. We use RF and XGB as base learners. The resulting CATE estimation models are denoted as `s_rf` and `s_xgb`.

2. *T-learners* (4 models). A T-learner (Künzel et al., 2019) constructs two separate models: one to estimate $\mu_1(\mathbf{x})$ using the treated units with $Z_i = 1$, and another to estimate $\mu_0(\mathbf{x})$ using the control units with $Z_i = 0$. Let $\hat{\mu}_1(\mathbf{x})$ and $\hat{\mu}_0(\mathbf{x})$ denote the resulting estimators. The CATE for an individual with features $\mathbf{x}_i$ is then estimated as $\hat{\tau}(\mathbf{x}_i) = \hat{\mu}_1(\mathbf{x}_i) - \hat{\mu}_0(\mathbf{x}_i)$. We use Lasso, logistic regression, RF and XGB as base learners. The resulting CATE estimation models are denoted as `t_lasso`, `t_logistic`, `t_rf` and `t_xgb`.

3. *X-learners* (4 models). An X-learner (Künzel et al., 2019) estimates CATE in three steps. First, it uses a T-learner to obtain the estimated response functions $\hat{\mu}_1(\mathbf{x})$ and $\hat{\mu}_0(\mathbf{x})$. Second, it imputes the treatment effects: for treated units with $Z_i = 1$, $\tilde{D}_i = Y_i - \hat{\mu}_0(\mathbf{x}_i)$;

for control units with $Z_i = 0$, $\tilde{D}_i = \hat{\mu}_1(\mathbf{x}_i) - Y_i$. It then trains two models: one on the treated units to predict $\tilde{D}_i$ and obtain $\hat{\tau}_1(\mathbf{x})$, and another on the control units to predict $\tilde{D}_i$ and obtain $\hat{\tau}_0(\mathbf{x})$. Third, the final CATE estimate is defined as a weighted average of $\hat{\tau}_1(\mathbf{x})$ and $\hat{\tau}_0(\mathbf{x})$. For the first step, we use Lasso, logistic regression, RF and XGB as base learners; for the second step, we use Lasso as the base learner. The resulting CATE estimation models are denoted as `x_lasso`, `x_logistic`, `x_rf` and `x_xgb`.

4. *R-learners* (9 models). A R-learner (Nie and Wager, 2021) estimates CATE in two steps. First, it estimates the conditional expectation of the outcome, $m(\mathbf{x}) = E[Y|\mathbf{X} = \mathbf{x}]$, and the conditional expectation of the treatment indicator, $e(\mathbf{x}) = E[Z|\mathbf{X} = \mathbf{x}]$. Let the resulting estimators be $\hat{m}(\mathbf{x})$ and $\hat{e}(\mathbf{x})$. Second, it estimates CATE by performing a weighted regression of the outcome residual $Y_i - \hat{m}(\mathbf{x}_i)$ on the treatment residual $Z_i - \hat{e}(\mathbf{x}_i)$. We use Lasso, RF and XGB as base learners in both steps. Unlike Dwivedi et al. (2020), who randomly select only four pairs of base learners, we use all nine pairs of base learners. The resulting estimation models are denoted as `r_lassolasso`, `r_lassorrf`, `r_lassoxgb`, `r_rflasso`, `r_rfrf`, `r_rfxgb`, `r_xgblasso`, `r_xgbrf`, `r_xgbxgb`.

5. *Causal Tree and Causal Forest* (4 models). A causal tree (Athey and Imbens, 2016) recursively partitions the feature space into subgroups that differ in the magnitude of their treatment effects. An honest approach is used for estimation, with one sample being used to construct the partition and another being used to estimate treatment effects for each subgroup. A causal forest (Wager and Athey, 2018) generates an ensemble of trees, each of which is built using random subsamples of the training set and outputs a CATE estimate. The forest then averages these estimates to get the final estimate. We use two versions

of the causal tree, with minimum leaf sizes of 500 and 2,000 observations, denoted by `causal_tree_1` and `causal_tree_2`. The two causal-forest versions use the same minimum leaf sizes and are denoted by `causal_forest_1` and `causal_forest_2`.

We tune all estimation models based on meta-learners via 4-fold cross validation using `cvorig`. For each type of base learner, we use a common hyperparameter grid and evaluate 200 random combinations of grid values each time. The hyperparameter search spaces cover the regularization strength for Lasso; the regularization type and strength for logistic regression; the minimum samples per leaf, maximum tree depth, number of trees, and bootstrap usage for RF; and the learning rate, number of boosting iterations, and regularization parameters for XGB, among others.

4 Calibration-Based Model Check

Since the true CATE $\tau(\mathbf{x}_i)$ is unobserved for any unit, standard model selection criteria, such as the area under the ROC curve (AUC), cannot be directly applied for model evaluation and selection. Consequently, we follow Dwivedi et al. (2020) and adopt a calibration-based approach to assess the accuracy and stability of the CATE estimation models.

The key idea is to notice that the average treatment effect for a subgroup of the feature space, $\mathbf{G}$, can be written as

$$E[Y(1) - Y(0)|\mathbf{X} \in \mathbf{G}] = E\{E[Y(1) - Y(0)|\mathbf{X}]\mathbf{X} \in \mathbf{G}\} = E[\tau(\mathbf{X})|\mathbf{X} \in \mathbf{G}]. \quad (1)$$

We bin observations by their CATE estimates and treat each bin as a subgroup. The rightmost part of (1) indicates that the subgroup average treatment effect can be estimated

by the bin-specific average of the CATE estimates. The leftmost part of (1) indicates that this effect can also be estimated by the difference-in-means within the subgroup. By comparing these two sets of estimates, we can assess the calibration of the CATE models.

4.1 The $\mathcal{R}_C^2$ Score

For a given CATE estimation model $\mathfrak{M}$, let $\mathbf{m}_q$ denote the q quantile of the model CATE estimates from a training fold. Using a fixed grid of quantile levels $q \in \{0.1, 0.2, \dots, 0.9\}$, we define the bin boundaries as:

$$\mathbf{m}_0 = -\infty < \mathbf{m}_{0.1} \leq \mathbf{m}_{0.2} \leq \dots \leq \mathbf{m}_{0.9} < \mathbf{m}_1 = \infty.$$

Let $\mathcal{G}_j$ ($j = 1, \dots, 10$) be the subset of the feature space where the model CATE estimate falls in the interval $[\mathbf{m}_{0.1(j-1)}, \mathbf{m}_{0.1j}]$.

For any set of observations $\mathcal{S}$, $\mathcal{G}_j \cap \mathcal{S}$ is the subset of observations belonging to the j -th subgroup, $\mathcal{G}_j$. Let $\bar{\mathfrak{m}}_{\mathcal{G}_j \cap \mathcal{S}}$ be the average model CATE estimate within this subset, and let $\hat{\tau}_{\mathcal{G}_j \cap \mathcal{S}}$ be the difference-in-means estimate within this subset. The calibration score measures the weighted average discrepancy between these two sets of estimates:

$$\text{Cal-Score}(\mathcal{S}; \mathfrak{M}) = \sum_{j=1}^{10} \frac{|\mathcal{G}_j \cap \mathcal{S}|}{|\mathcal{S}|} \cdot |\bar{\mathfrak{m}}_{\mathcal{G}_j \cap \mathcal{S}} - \hat{\tau}_{\mathcal{G}_j \cap \mathcal{S}}| \quad (2)$$

Let $\hat{\tau}_{\text{ATE}}$ denote the difference-in-means estimate for all units in $\mathcal{S}$. The baseline calibration score is defined by replacing the average model CATE estimate in (2) with this constant:

$$\text{Cal-Score}(\mathcal{S}; \hat{\tau}_{\text{ATE}}) = \sum_{j=1}^{10} \frac{|\mathcal{G}_j \cap \mathcal{S}|}{|\mathcal{S}|} \cdot |\hat{\tau}_{\text{ATE}} - \hat{\tau}_{\mathcal{G}_j \cap \mathcal{S}}|$$

The $\mathcal{R}_C^2(\mathcal{S}; \mathfrak{M})$ score normalizes the calibration score for model $\mathfrak{M}$ relative to this baseline:

$$\mathcal{R}_C^2(\mathcal{S}; \mathfrak{M}) = 1 - \frac{\text{Cal-Score}(\mathcal{S}; \mathfrak{M})}{\text{Cal-Score}(\mathcal{S}; \hat{\tau}_{\text{ATE}})}.$$

This score can take any value in $(-\infty, 1]$. A value close to 1 indicates good calibration, while a value close to zero or negative suggests that the model $\mathfrak{M}$ does no better than a constant ATE estimate at capturing heterogeneity, indicating potential overfitting.

4.2 Poor Global Calibration

Figure 2 shows the histogram of $\mathcal{R}_C^2$ -scores across all 23 CATE estimation models. The training fold distribution (left panel) concentrates around positive values with a mean of 0.30 and a median of 0.28. In contrast, the validation fold distribution (right panel) concentrates around negative values with a mean of -0.21 and a median of -0.13 . This systematic performance gap is characteristic of overfitting, indicating that the models are fitting noise in the training data rather than genuine treatment effect heterogeneity.

Figure 3 presents calibration plots for two models: the X-learner with RF as the base learner (`x_rf`), and the R-learner which uses RF in the first step and Lasso in the second step (`r_rflasso`). The plots show the average model CATE estimate, $\bar{\mathfrak{M}}_{\mathcal{G}_j \cap \mathcal{S}}$, versus the corresponding difference-in-means estimate, $\hat{\tau}_{\mathcal{G}_j \cap \mathcal{S}}$, across ten quantile-based bins for one training-validation fold pair, with error bars indicating ± 1 standard error. The horizontal dashed line indicates the overall ATE estimate of -0.0158 , computed from all 28,819 customers.

On the training folds (left panels), both models show reasonable calibration, with the model estimates closely tracking the difference-in-means estimates across bins. In contrast,

calibration deteriorates markedly on the validation folds (right panels) where the difference-in-means estimates exhibit high variance and frequently deviate from the monotonic trend in the model estimates.

Our result echoes the finding from the original StaDISC analysis of a clinical trial dataset in Dwivedi et al. (2020), in which even advanced machine learning methods led to similarly poor global calibration.

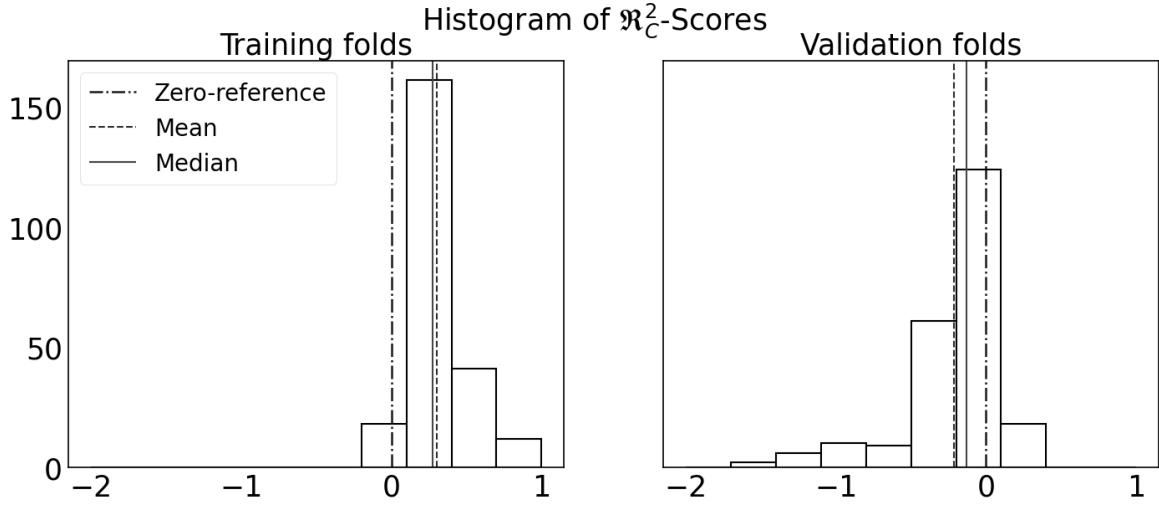


Figure 2: Histogram of $\mathcal{R}_C^2$ -scores on training versus validation folds across 12 training-validation fold pairs for 23 CATE estimation models.

4.3 The Quantile-based Top Subgroups

Although the CATE models calibrate poorly globally, it is possible that some models may identify subgroups with treatment effects higher than average. For $q \in (0, 1)$, let $\mathcal{L}_q$ denote the subset of the feature space where the model CATE estimate is strictly below $\mathbf{m}_q$. Its complement, $\mathcal{U}_q$, contains scores greater than or equal to $\mathbf{m}_q$ and is called a quantile-based top subgroup. Scores tied at the cutoff are assigned to the top subgroup, so its realized

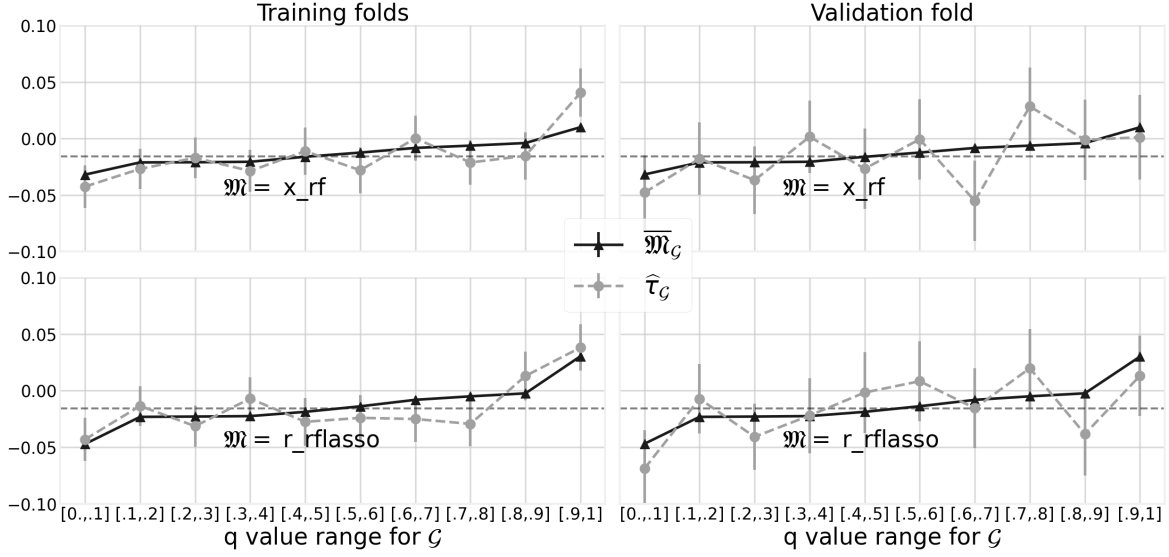


Figure 3: Calibration plots for two models: `x_rf` and `r_rflasso`. Each panel compares the average model CATE estimates (triangles) with the difference-in-means estimates (circles) across ten quantile-based bins, with error bars indicating ± 1 standard error. The left column shows results on the training folds, and the right column shows those on the validation folds. The horizontal dashed line marks the difference-in-means estimate of ATE for all 28,819 customers.

share in a validation or holdout sample can differ from the nominal fraction. Different models and training folds lead to different partitions of the feature space.

For a validation fold $\mathcal{S}_{VF}$ and for $q \in \{0.9, 0.8, 0.7, 0.6, 0.5\}$, we perform a directional check by investigating whether the difference-in-means estimate for units in the quantile-based top subgroup, $\mathcal{U}_q \cap \mathcal{S}_{VF}$, is larger than that for units not in the quantile-based top subgroup, $\mathcal{L}_q \cap \mathcal{S}_{VF}$. Define a binary indicator

$$B_q = I \left(\hat{\tau}_{\mathcal{L}_q \cap \mathcal{S}_{VF}} < \hat{\tau}_{\mathcal{U}_q \cap \mathcal{S}_{VF}} \right).$$

A B_q value of 1 indicates that the estimated top-subgroup effect exceeds the estimated complementary-subgroup effect in that validation fold. Let $\bar{B}_q$ be the average of B_q across finite validation-fold contrasts. It is a directional sign frequency: a value close to 1 indicates consistent top-versus-rest ordering, but does not measure the magnitude or global

calibration of the CATE estimates.

Figure 4 shows box plots of $\bar{B}_q$ ($q \in \{0.9, 0.8, 0.7, 0.6, 0.5\}$) across all 23 models. Overall, most models exhibit poor calibration and instability. This finding contrasts with the original StaDISC analysis of a clinical trial dataset in Dwivedi et al. (2020), in which the box plots of $\bar{B}_q$ are all concentrated around values larger than 0.8, with medians being 0.9. Relatively speaking, the results for $q = 0.9$ show the best generalization performance. Therefore, in the next section, we use the calibration plots not to choose one fixed estimator, but to order estimators by validation-fold calibration slope and then select a top-10% targeting ensemble along the resulting nested path.

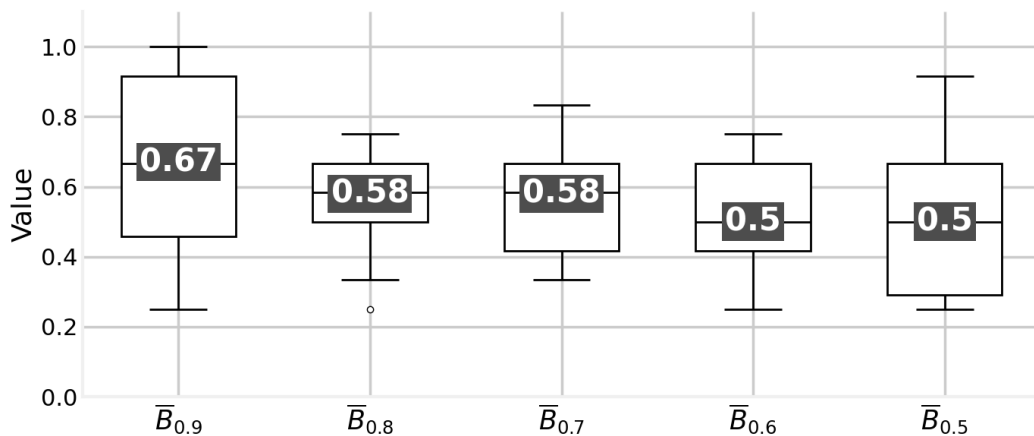


Figure 4: Box plots of $\bar{B}_q$ ($q \in \{0.9, 0.8, 0.7, 0.6, 0.5\}$) across all 23 CATE estimation models.

5 Model Selection and Validation

5.1 Model Selection Procedure

We use the validation-fold calibration slope, CalSlope, to rank candidate estimators. Let $\mathcal{T}_f$ and $\mathcal{V}_f$ denote the training and validation parts of fold f , respectively, and let $\hat{\tau}_{im}^{(-f)}$ be the

CATE prediction for unit i from estimator m trained without the validation fold. For each estimator-fold pair (m, f) , we form five CATE strata by cutting the training-fold predictions $\{\hat{\tau}_{im}^{(-f)} : i \in \mathcal{T}_f\}$ at their empirical quintiles (the 20th, 40th, 60th, and 80th percentiles of the training-fold CATE distribution) and applying those cutoffs to the validation fold. This separation ensures that the stratum boundaries are determined without using validation outcomes. Denote the resulting validation strata by $\mathcal{H}_{mf1}, \dots, \mathcal{H}_{mf5}$, ordered from lowest to highest predicted CATE.

For each nonempty stratum $\mathcal{H}_{mfk}$, define its mean predicted CATE and observed treatment effect as

$$\bar{\mu}_{mfk} = \frac{1}{|\mathcal{H}_{mfk}|} \sum_{i \in \mathcal{H}_{mfk}} \hat{\tau}_{im}^{(-f)}, \quad \hat{\theta}_{mfk} = \frac{1}{n_{1,mfk}} \sum_{\substack{i \in \mathcal{H}_{mfk} \\ Z_i=1}} Y_i - \frac{1}{n_{0,mfk}} \sum_{\substack{i \in \mathcal{H}_{mfk} \\ Z_i=0}} Y_i,$$

where $n_{z,mfk} = \sum_{i \in \mathcal{H}_{mfk}} I(Z_i = z)$. The fold-level calibration slope is the ordinary least-squares slope in the regression of the observed stratum ATE on the mean predicted CATE:

$$(\hat{\alpha}_{mf}, \hat{\beta}_{mf}) = \arg \min_{\alpha, \beta} \sum_{k \in \mathcal{K}_{mf}} \left(\hat{\theta}_{mfk} - \alpha - \beta \bar{\mu}_{mfk} \right)^2, \quad \text{CalSlope}_{mf} = \hat{\beta}_{mf},$$

where $\mathcal{K}_{mf}$ is the set of valid strata. We compute this slope only when at least two valid strata remain and the $\bar{\mu}_{mfk}$ values are not all equal. The estimator-level statistic is the average over valid validation folds,

$$\text{CalSlope}_m = \frac{1}{|\mathcal{F}_m|} \sum_{f \in \mathcal{F}_m} \text{CalSlope}_{mf}.$$

A large positive value means that higher predicted-CATE strata have higher realized treatment effects on validation data; a value near zero indicates little ranking signal; and a negative value indicates that the validation ranking is reversed. The estimators are ordered from largest to smallest CalSlope_m .

This ordered list defines a nested path of equal-weight mean ensembles. The first ensemble contains the top-ranked estimator alone; the second contains the top two estimators; and so on until the full candidate set is included. For ensemble e in fold f , let $\widehat{s}_{ie}^{(-f)}$ be the average of the CATE predictions from its member estimators. We compute the top-10% cutoff from $\{\widehat{s}_{ie}^{(-f)} : i \in \mathcal{T}_f\}$ and apply that cutoff to $\mathcal{V}_f$ to define the validation top subgroup $\mathcal{G}_{ef}^{0.10}$ and its complement $\mathcal{R}_{ef}^{0.10}$. The top subgroup includes scores tied at the training-fold 90th-percentile cutoff and therefore need not contain exactly 10% of the validation observations. The fold-level targeting contrast is

$$\Delta_{ef,0.10} = \widehat{\delta}(\mathcal{G}_{ef}^{0.10}) - \widehat{\delta}(\mathcal{R}_{ef}^{0.10}),$$

where $\widehat{\delta}(\cdot)$ is the Neyman difference-in-means estimator. We then average this contrast over validation folds,

$$\overline{\Delta}_{0.10}(e) = \frac{1}{|\mathcal{F}_e|} \sum_{f \in \mathcal{F}_e} \Delta_{ef,0.10}.$$

The selected ensemble is the nested-path candidate with the largest $\overline{\Delta}_{0.10}$. If two ensembles tie on $\overline{\Delta}_{0.10}$, we keep the one with fewer estimators. Thus the public selection rule uses only two statistics: CalSlope for ordering candidate estimators and $\overline{\Delta}_{0.10}$ for selecting one ensemble from the nested path.

5.2 Simulation Evaluation

We evaluate the selection procedure on two binary-outcome data-generating processes (DGPs) for which the ground-truth CATE is known. In each DGP,

$$X_i \sim N(0, \Sigma_d), \quad (\Sigma_d)_{jk} = \rho_d^{|j-k|}, \quad W_i \sim \text{Bernoulli}(0.5),$$

and the outcome is generated as

$$Y_i \mid X_i, W_i \sim \text{Bernoulli}[\Lambda\{b_d(X_i) + (W_i - 0.5)\eta_d(X_i)\}], \quad \Lambda(u) = \{1 + \exp(-u)\}^{-1}.$$

DGP1 uses $n = 12,000$, 50 covariates, $\rho_1 = 0.5$, and five active covariates, with

$$b_1(X) = \max\{0, X_1 + X_2, X_3\} + 0.4(X_4^2 - 1) + 0.3X_2X_5 + 0.4\max\{0, X_4 + X_5\} - 0.3,$$

$$\eta_1(X) = -0.05 + 0.15\{X_1 + \log(1 + \exp X_2)\}.$$

DGP2 uses $n = 8,000$, 75 covariates, $\rho_2 = 0.6$, and ten active covariates. Its nuisance surface is

$$\begin{aligned} b_2(X) = & 0.81\{X_1 + X_2 > 0\} - 0.61\{X_3 > 0.5\} + 0.51\{X_4X_5 > 0\} \\ & + 0.4\max(0, X_6 + X_7) + 0.3(X_8^2 - 1) - 0.2X_9X_{10} - 0.4, \end{aligned}$$

and its treatment-effect index is

$$\eta_2(X) = 0.02 + 0.10 \tanh(0.8X_1 + 0.6X_2 - 0.4X_3) + 0.06\Lambda(X_4 + X_5) + 0.04X_6.$$

The ground-truth CATE on the outcome-probability scale is therefore $\tau_d(X) = \Lambda\{b_d(X) + \eta_d(X)/2\} - \Lambda\{b_d(X) - \eta_d(X)/2\}$.

In each replication, we reserve 20% of observations as an independent holdout set. On the remaining 80%, we screen features by taking the union of the ten largest absolute Lasso coefficients fitted separately in the two treatment arms. Hyperparameters of the 19 meta-learners are tuned on the original four-fold partition, and those learners together with the four fixed causal-tree and causal-forest specifications are then fitted, yielding the same 23-estimator library used in the application. The selection statistics CalSlope

and $\bar{\Delta}_{0.10}$ are computed over 12 validation folds. After selection, the chosen members are refitted on the full training-validation sample, their equal-weight score determines the training-validation 90th percentile, and that threshold is transferred to the holdout scores. Ground-truth CATE is revealed only after selection and is averaged within the resulting holdout subgroup. Individual estimators are evaluated by the same transferred-threshold rule on the same usable replications.

The archived results contain 99 complete DGP1 replications and 100 complete DGP2 replications. The selected ensemble achieves a mean ground-truth top-10% holdout CATE of 0.0277 in DGP1 and 0.0710 in DGP2. Figure 5 ranks these means against those of all 23 individual estimators. In each DGP, the selected ensemble ranks second among the 24 displayed methods, and the leading individual estimators are the S-learner with a random-forest base learner in DGP1 and the X-learner with a random-forest base learner in DGP2. Supplementary Section S1.6 evaluates $\bar{B}_{0.9}$ as an alternative first-stage ordering score within the same nested-ensemble framework.

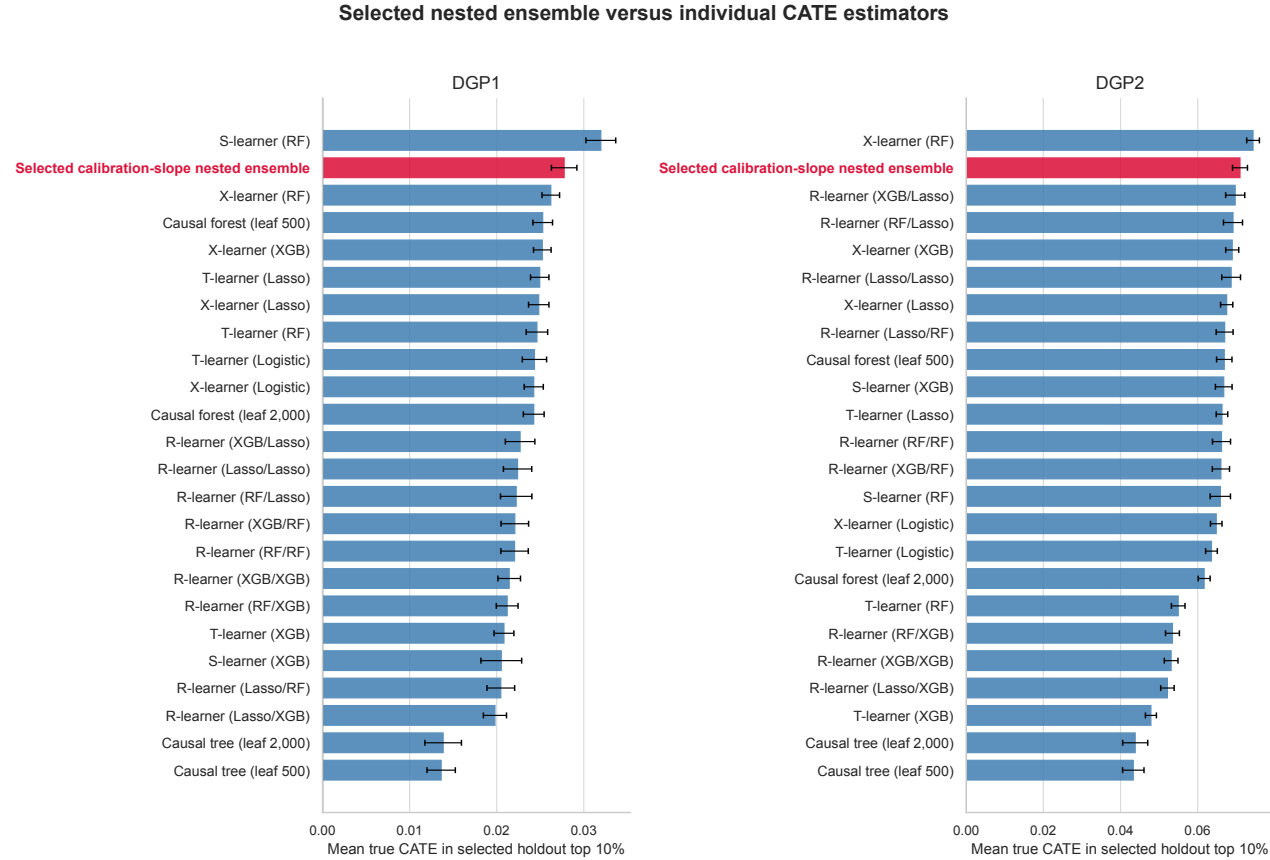


Figure 5: The selected calibration-slope nested ensemble compared with all 23 individual CATE estimators. Within each DGP, bars show the Monte Carlo mean ground-truth CATE above the transferred 90th-percentile holdout threshold; error bars equal 1.96 Monte Carlo standard errors of the mean. Selection does not use true CATE. The highlighted ensemble ranks second among the 24 displayed methods in both DGPs.

5.3 Final Validation of The Selected Model

We fit the selected ensemble on the full training-validation dataset, determine the top-10% cutoff, and apply the ensemble to the holdout test set. Table 2 summarizes the results. For the training-validation dataset, the top-10% subgroup yields an ATE of 0.0407 (SE = 0.0175), with a 95% confidence interval that excludes zero. For the holdout test set, the top-10% subgroup yields an ATE of 0.0617 (SE = 0.0346), with a 90% confidence interval that excludes zero. The holdout estimate is positive and nominally significant at the 10% level. It should not be interpreted as an unadjusted confirmatory p -value because the scoring-function development process included a holdout-assisted screening step.

Table 2: Final Validation Results (Main Treatment Arm)

	Train-Val	Holdout
Total sample size	28,819	7,205
Subgroup size	2,882	723
Subgroup ATE (SE)	0.0407 (0.0175)	0.0617 (0.0346)
t -statistic	2.32	1.78
95% CI	[0.0064, 0.0750]	[−0.0061, 0.1296]
90% CI	[0.0119, 0.0695]	[0.0048, 0.1187]

6 Conclusion

In behavioral finance experiments, responses to interventions are often heterogeneous and exhibit a low signal-to-noise ratio. Our extension of the StaDISC methodology to the Turkish bank overdraft experiment has demonstrated a nuanced picture of heterogeneous treatment effects, which can inform personalized interventions. On one hand, global CATE estimation fails: all models trained exhibit poor calibration on validation folds. On the

other hand, a calibration-slope nested-ensemble procedure identifies a quantile-based top subgroup with a holdout treatment effect of 0.0617, nominally significant at the 10% level. In the two simulation designs, the procedure ranks second among the 23 individual estimators plus the selected ensemble by mean true top-10% holdout CATE. Supplementary Material reports sensitivity to the first-stage ordering score and to alternative second-stage path-selection functions. The main simulation evidence supports the combined use of continuous calibration ordering, nested ensembling, and a targeting-aligned final contrast in the designs considered, while remaining exploratory rather than establishing universal dominance. Our results connect to Yadlowsky et al. (2025)’s observation that treatment prioritization rules can add value even when CATE estimates are miscalibrated, and to Fernández-Loría and Provost (2022)’s theoretical analysis showing that ranking-based targeting requires weaker conditions than accurate effect prediction.

The bill-payment bundled message is the only treatment arm that yields a positive holdout effect under the procedure. The debit-card bundled message and the availability-reminder comparison do not produce positive holdout effects (Supplementary Section S2). The development audit examined a broader set of targeting summaries, documented in Supplementary Section S1.7. Because the final criterion was chosen after consulting both simulation truth and real-data holdout diagnostics, that audit documents sensitivity and method development rather than independent corroboration.

Relative to the original StaDISC application in clinical trials (Dwivedi et al., 2020), our setting has smaller average treatment effects and noisier outcomes, yet the similar steps and components—calibration-based reality checks, model comparison driven by validation-fold

performance, and evaluation using a quantile-based top subgroup—still deliver a useful targeting rule. This suggests that the underlying PCS framework can serve as a general template for analyzing heterogeneous treatment effects in behavioral finance experiments. The PCS framework’s emphasis on stability proves essential for separating genuine heterogeneity from statistical artifacts. Several models in our analysis achieve impressive performance on particular cross-validation splits but perform poorly when the split changes. This supports the broader principle that conclusions robust to reasonable perturbations deserve greater attention in practice.

The original StaDISC methodology includes a CellSearch step to translate quantile-based subgroups into interpretable feature combinations. To reduce the search space in this step, the features are discretized and the number of features is limited. Since we omit this step, we cannot provide simple targeting rules (e.g., “customers with balance greater than X and tenure less than Y”). Instead, a bank would need to deploy the selected model to estimate CATE for each customer. However, unlike the original approach, our models can use continuous features and accommodate a larger number of them. As the deployment of machine learning models becomes standard practice in many financial institutions, we do not view this omission as a substantive limitation.

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Disclosure Statement

The authors report there are no competing interests to declare.

Data Availability

The data underlying this article are available from the original study by Alan et al. (2018).

Replication code is available from the authors upon request.

References

- S. Alan, M. Cevalcar, D. Karlan, and J. Zinman. Unshrouding: Evidence from bank overdrafts in turkey. *The Journal of Finance*, 73(2):481–522, 2018.
- E. Ascarza. Retention futility: Targeting high-risk customers might be ineffective. *Journal of Marketing Research*, 55(1):80–98, 2018.
- S. Athey and G. Imbens. Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27):7353–7360, 2016.
- S. Athey, N. Keleher, and J. Spiess. Machine learning who to nudge: Causal vs predictive targeting in a field experiment on student financial aid renewal. *Journal of Econometrics*, 249(1):105945, 2025.
- M. Bertrand, D. Karlan, S. Mullainathan, E. Shafir, and J. Zinman. What’s advertising content worth? evidence from a consumer credit marketing field experiment. *The Quarterly Journal of Economics*, 125(1):263–306, 2010.
- R. Dwivedi, Y. S. Tan, B. Park, M. Wei, K. Horgan, D. Madigan, and B. Yu. Stable discovery of interpretable subgroups via calibration in causal studies. *International Statistical Review*, 88(S1):S135–S178, 2020.
- C. Fernández-Loría and F. Provost. Causal classification: Treatment effect estimation vs outcome prediction. *The Journal of Machine Learning Research*, 23(59):1–35, 2022.
- X. Gabaix and D. Laibson. Shrouded attributes, consumer myopia, and information sup-

- pression in competitive markets. *The Quarterly Journal of Economics*, 121(2):505–540, 2006.
- C. Huber, N. Benda, and T. Friede. A comparison of subgroup identification methods in clinical drug development: Simulation study and regulatory considerations. *Pharmaceutical Statistics*, 18(5):600–626, 2019.
- D. Karlan, M. McConnell, S. Mullainathan, and J. Zinman. Getting to the top of mind: How reminders increase saving. *Management Science*, 62(12):3393–3411, 2016.
- S. R. Künnel, J. S. Sekhon, P. J. Bickel, and B. Yu. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165, 2019.
- X. Nie and S. Wager. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319, 2021.
- T. Ondra, A. Dmitrienko, T. Friede, A. Graf, F. Miller, N. Stallard, and M. Posch. Methods for identification and confirmation of targeted subgroups in clinical trials: A systematic review. *Journal of Biopharmaceutical Statistics*, 26(1):99–119, 2016.
- S. Wager and S. Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018.
- S. Yadlowsky, S. Fleming, N. Shah, E. Brunskill, and S. Wager. Evaluating treatment

prioritization rules via rank-weighted average treatment effects. *Journal of the American Statistical Association*, 120(549):38–51, 2025.

B. Yu and K. Kumbier. Three principles of data science: Predictability, computability, and stability (PCS). In *2018 IEEE International Conference on Big Data (Big Data)*, pages 3920–3929, 2018.

B. Yu and K. Kumbier. Veridical data science. *Proceedings of the National Academy of Sciences*, 117(8):3920–3929, 2020.

Supplementary Material

S1 Scoring-Function Development and Evaluation

S1.1 Scope and AI-assisted generation

Each of the 40 second-stage path-selection functions maps fold-level predictions, treatment assignments, and observed outcomes to a scalar that ranks nested equal-weight ensemble candidates on a path already defined by the first-stage ordering score. The candidate with the largest value is selected, and the deployment target is fixed at the predicted top 10% subgroup.

Candidates were generated with assistance from OpenAI GPT-5.5 (accessed July 14, 2026), screened for computability, redundancy, and alignment with the targeting objective, and translated into executable code. The language model was not part of the executable pipeline.

Table S1 groups the second-stage functions by purpose. The stability-penalized top-decile mean is heuristic rather than an inferential confidence bound because the repeated cross-validation contrasts are dependent.

S1.2 Fold-level data objects

For each analysis, three stratified four-fold partitions were used: the original partition and two perturbed partitions. Thus $\mathcal{F}$ contains 12 training-validation fold pairs. For candidate ensemble e and fold f , member-model predictions were averaged arithmetically to obtain

s_{ief} for unit i . For each selected fraction

$$Q = \{0.10, 0.20, 0.30, 0.40, 0.50\},$$

the cutoff was the empirical $(1-q)$ quantile of s_{ief} in the training portion $\mathcal{T}_f$. Applying that training-derived cutoff to $\mathcal{V}_f$ formed the validation top subgroup G_{efq} and its complement R_{efq} . No validation outcomes were used to form a cutoff.

For any subgroup A , let

$$\widehat{\delta}(A) = \frac{\sum_{i \in A} Z_i Y_i}{\sum_{i \in A} Z_i} - \frac{\sum_{i \in A} (1 - Z_i) Y_i}{\sum_{i \in A} (1 - Z_i)}$$

be the Neyman difference-in-means estimator. The common targeting quantities were

$$\Delta_{efq} = \widehat{\delta}(G_{efq}) - \widehat{\delta}(R_{efq}), \tag{S1}$$

$$B_{efq} = \mathbf{1}\{\Delta_{efq} > 0\}, \tag{S2}$$

$$t_{efq} = \Delta_{efq} / \widehat{\text{se}}(\Delta_{efq}), \tag{S3}$$

where the standard error of the contrast was the square root of the sum of the two subgroup variance estimates. A fold-fraction value was set to missing when a required treated or control cell was empty; aggregations used the available finite values.

Calibration quantities used five strata. The 20th, 40th, 60th, and 80th percentiles of the training-fold score distribution were applied to validation scores. For stratum b , $\bar{\mu}_{efb}$ denotes mean predicted CATE and $\widehat{\theta}_{efb}$ denotes the validation difference-in-means estimate. Empty treatment cells were omitted, and calibration-line quantities required at least two valid strata with nonconstant $\bar{\mu}_{efb}$.

The targeting-curve summaries used ten equally spaced fractions from 0.02 through 0.10. At each fraction, validation units were sorted by predicted CATE, the corresponding top

group was formed, and its ATE was calculated. The implemented **rate** statistic averages these top-group ATEs; **qini** averages their differences from the full validation-fold ATE. Scores labeled **lcb** use the multiplier 1.64485 as a stability penalty. They are not confidence intervals for the cross-fold mean because the repeated fold contrasts are dependent.

S1.3 Notation for the second-stage function table

Table S1 uses finite-value aggregation throughout. Let $N_{\Delta,e}$ denote the number of finite Δ_{efq} values over (f, q) and let $N_{\Delta,e,0.10}$ denote the number of finite top-decile fold contrasts. Bars denote arithmetic means over those finite values, $s_{\Delta,e}$ and $s_{\Delta,e,0.10}$ are their sample standard deviations, and $\bar{\Delta}_e(q)$ is the fold mean at fixed q . The quantity $\bar{B}_e$ is the finite-value mean of B_{efq} over folds and fractions.

For calibration scores, $\hat{\theta}_{efb}^{\text{fit}}$ is the fitted value from the fold-specific OLS calibration line evaluated at $\bar{\mu}_{efb}$. For a candidate-path quantity v_e , $R_e(v)$ is its ascending within-path percentile rank with average ranks for ties, so larger values receive larger ranks. The sign-consistency factor is $A_e^{\text{sign}} = \max\{\bar{B}_e, 1 - \bar{B}_e\}$, S_e^{IQR} is the mean validation-fold CATE-score interquartile range, and $\varepsilon = 10^{-12}$. In the trimmed-mean definition, $\Delta_{e(j),0.10}$ is the j th order statistic of the finite top-decile fold contrasts and $N = N_{\Delta,e,0.10}$.

S1.4 Candidate path and deterministic selection

The candidate library contains 23 estimators: 19 meta-learners, two causal trees, and two causal forests. All 23 are present in the simulation caches and in each of the three saved real-data perturbation libraries. Within each realization, the estimator with the largest

mean validation-fold calibration slope was designated as the anchor. The remaining 22 estimators were ordered from largest to smallest calibration slope. This ordering generated 23 nested candidates: the anchor alone, the anchor plus the first ordered estimator, and so forth through the arithmetic mean of all 23 estimators.

Each of the 40 second-stage functions in Table S1 was computed separately for every candidate on this same path. For a fixed function, the candidate with the largest finite value was selected. Exact ties were resolved first in favor of fewer estimators and then by the stored candidate name, making the map from fold predictions to a selected ensemble deterministic. Intermediate components were computed only where required by the documented composite formulas.

These are alternative criteria for development, not simultaneous requirements imposed by the final procedure.

Table S1: The 40 distinct second-stage path-selection functions evaluated during development. Each rank compares the selected ensemble’s Monte Carlo mean ground-truth holdout-subgroup CATE with the same 23 individual estimators; rank 1 is best of 24.

Statistic	Definition	Meaning	DGP1 rank	DGP2 rank
weighted_b	$W_e = \frac{\sum_{f,q} \max\{\Delta_{efq}, 0\}}{\sum_{f,q} \Delta_{efq} }$	Share of absolute targeting-contrast mass that is positive.	2	2
mean_delta	$\bar{\Delta}_e = \frac{1}{ \mathcal{F} Q } \sum_{f,q} \Delta_{efq}$	Average top- q -minus-rest treatment-effect contrast over Q .	3	2
mean_delta_top10	$\bar{\Delta}_{e,0.10} = \frac{1}{ \mathcal{F} } \sum_f \Delta_{ef,0.10}$	Top-10%-minus-rest validation contrast.	2	2
lcb_delta_top10	$\text{LCB}_{\Delta,e,0.10} = \bar{\Delta}_{e,0.10} - z_{0.95} s_{\Delta,e,0.10} / \sqrt{N_{\Delta,e,0.10}}$	Heuristic stability-penalized top-10%-minus-rest contrast. It is not a confidence bound because fold contrasts are dependent.	2	2
median_delta_top10	$\text{Med}_{\Delta,e,0.10} = \text{median}_f \Delta_{ef,0.10}$	Median finite fold-level top-10%-minus-rest contrast.	2	2
trimmed_mean_delta_top10	$\text{Trim}_{\Delta,e,0.10} = (N - 2k)^{-1} \sum_{j=k+1}^{N-k} \Delta_{e(j),0.10}, k = \lfloor 0.1N \rfloor$	Symmetric 10% trimmed mean of the ordered finite fold-level top-10% contrasts.	2	2
epm	$\text{EPM}_e = \frac{1}{ \mathcal{F} Q } \sum_{f,q} \max\{\Delta_{efq}, 0\}$	Expected positive margin of the targeting contrast.	3	2
snr	$\text{SNR}_e = \bar{\Delta}_e / s_{\Delta,e}$	Signal-to-noise ratio of top- q -minus-rest contrasts.	2	2
lcb_delta	$\text{LCB}_{\Delta,e} = \bar{\Delta}_e - z_{0.95} s_{\Delta,e} / \sqrt{N_{\Delta,e}}$	Heuristic normal-penalized mean contrast across dependent folds and fractions.	3	2
top_lcb	$\frac{1}{ \mathcal{F} Q } \sum_{f,q} \{\hat{\tau}(G_{efq}) - z_{0.95} \text{se}[\hat{\tau}(G_{efq})]\}$	Average normal-penalized top-group ATE score.	2	2
rate	$\text{RATE}_e = \frac{1}{ \mathcal{F} H } \sum_{f,h \in H} \hat{\tau}(G_{efh})$	Average top-ATE curve over ten equally spaced h values from 0.02 to 0.10.	2	2
qini	$\text{Qini}_e = \frac{1}{ \mathcal{F} H } \sum_{f,h \in H} \{\hat{\tau}(G_{efh}) - \hat{\tau}(V_f)\}$	Average uplift curve over ten equally spaced h values from 0.02 to 0.10.	2	2
cal_slope	$\text{CalSlope}_e = \frac{1}{ \mathcal{F} } \sum_f \frac{\text{Cov}_b(x_{efb}, a_{efb})}{\text{Var}_b(x_{efb})}$	Candidate-ensemble calibration slope across five predicted-CATE strata; distinct from estimator-level first-stage CalSlope_m .	2	6
cal_r2	$\frac{1}{ \mathcal{F} } \sum_f \left[1 - \frac{\sum_b (a_{efb} - \hat{a}_{efb})^2}{\sum_b (a_{efb} - \hat{a}_{ef})^2} \right]$	Linear calibration R^2 across the five CATE strata.	3	2
cal_monotonicity	$\frac{1}{ \mathcal{F} } \sum_f \binom{5}{2}^{-1} \sum_{b < c} \mathbf{1}\{a_{efb} \leq a_{efc}\}$	Fraction of stratum ATE pairs monotone increasing in predicted CATE.	3	2
neg_cal_rmse	$-\frac{1}{ \mathcal{F} } \sum_f \sqrt{5^{-1} \sum_b (a_{efb} - \hat{a}_{efb})^2}$	Negative calibration-line RMSE.	2	2
strata_ate_sd	$\frac{1}{ \mathcal{F} } \sum_f \text{sd}_b(a_{efb})$	Dispersion of realized stratum ATEs across calibration strata.	3	2
top_bin_is_best	$\frac{1}{ \mathcal{F} } \sum_f \mathbf{1}\{a_{ef,5} \geq \max_b a_{efb}\}$	Frequency that the highest predicted-CATE bin has the largest realized ATE.	2	2
neg_targeting_slope	$-\frac{\text{Cov}_{q \in Q}\{q, \bar{\Delta}_e(q)\}}{\text{Var}_{q \in Q}(q)}$	Negative slope of average targeting contrast versus selected fraction q .	2	3
score_iqr_neg	$- \mathcal{F} ^{-1} \sum_f \text{IQR}\{s_{ief} : i \in V_f\}$	Negative validation CATE-score interquartile range.	2	7

Continued on next page

Table S1 (continued)

Statistic	Definition	Meaning	DGP1 rank	DGP2 rank
score_sd_neg	$- \mathcal{F} ^{-1} \sum_f \text{sd}\{s_{ief} : i \in V_f\}$	Negative validation CATE-score standard deviation.	2	6
score_mad_neg	$- \mathcal{F} ^{-1} \sum_f \text{median}_{i \in V_f} s_{ief} - \text{median}_{j \in V_f} s_{jef} $	Negative validation CATE-score median absolute deviation.	2	7
score_p90_p10_neg	$- \mathcal{F} ^{-1} \sum_f \{P_{90}(s_{ef}) - P_{10}(s_{ef})\}$	Negative 90–10 spread of validation CATE scores.	2	6
score_range_neg	$- \mathcal{F} ^{-1} \sum_f \{\max_{i \in V_f} s_{ief} - \min_{i \in V_f} s_{ief}\}$	Negative validation CATE-score range.	2	7
top_tail_ratio	$ \mathcal{F} ^{-1} \sum_f \frac{\text{mean}\{s_{ief} : i \in G_{ef,0.10}\}}{ \text{mean}\{s_{ief} : i \in V_f\} }$	Ratio of top-10% mean predicted CATE to absolute overall validation mean predicted CATE.	2	7
bbar_epm	$S_e = \bar{B}_e \text{EPM}_e$	Product of sign frequency and expected positive margin.	3	2
bbar_delta	$S_e = \bar{B}_e \bar{\Delta}_e$	Product of sign frequency and mean contrast.	3	2
weighted_b_epm	$S_e = W_e \text{EPM}_e$	Product of weighted positive share and expected positive margin.	3	2
positive_cal_epm	$S_e = \max\{\text{CalSlope}_e, 0\} \text{EPM}_e$	Positive calibration slope multiplied by expected positive margin.	2	5
positive_lcb_epm	$S_e = \max\{\text{LCB}_{\Delta,e}, 0\} \text{EPM}_e$	Positive normal-penalized contrast multiplied by expected positive margin.	3	2
positive_cal_bbar_epm	$S_e = \max\{\text{CalSlope}_e, 0\} \bar{B}_e \text{EPM}_e$	Positive calibration slope, sign frequency, and positive margin product.	2	6
rank_bbar_epm	$S_e = \{R_e(\bar{B}) + R_e(\text{EPM})\}/2$	Mean within-path percentile rank of $\bar{B}$ and EPM.	3	2
rank_bbar_delta	$S_e = \{R_e(\bar{B}) + R_e(\bar{\Delta})\}/2$	Mean within-path percentile rank of $\bar{B}$ and mean contrast.	3	2
rank_bbar_cal_epm	$S_e = \{R_e(\bar{B}) + R_e(\text{CalSlope}) + R_e(\text{EPM})\}/3$	Mean within-path percentile rank of sign frequency, calibration slope, and EPM.	2	2
maximin_bbar_epm	$S_e = \min\{R_e(\bar{B}), R_e(\text{EPM})\}$	Maximin within-path percentile rank over sign frequency and EPM.	3	2
cal_slope_x_delta	$S_e = \max\{\text{CalSlope}_e, 0\} \bar{\Delta}_e$	Product of positive calibration slope and mean contrast.	2	5
smooth_delta_ratio	$S_e = \bar{\Delta}_e / (S_e^{\text{IQR}} + \varepsilon)$	Mean contrast divided by CATE-score IQR.	2	7
smooth_lcb_ratio	$S_e = \max\{\text{LCB}_{\Delta,e}, 0\} / (S_e^{\text{IQR}} + \varepsilon)$	Positive normal-penalized contrast divided by CATE-score IQR.	2	6
top_balance_delta	$S_e = A_e^{\text{sign}} \bar{\Delta}_e$	Mean contrast weighted by sign consistency.	3	2
stability_x_lcb	$S_e = A_e^{\text{sign}} \max\{\text{LCB}_{\Delta,e}, 0\}$	Positive normal-penalized contrast weighted by sign consistency.	3	2

S1.5 Simulation evaluation

The second-stage functions were evaluated using 99 DGP1 and 100 DGP2 Monte Carlo replications. Each replication contained fold predictions from all 23 estimators under the three four-fold perturbations, full training-validation and holdout predictions, observed outcomes and treatment assignments, and the simulation’s true CATE. For each function and replication, selection used only fold predictions, treatment assignments, and observed training-validation outcomes. The saved full-training-validation predictions of the selected members were averaged, the corresponding 90th-percentile cutoff was transferred to the saved holdout scores, and the true mean CATE of the selected holdout subgroup was calculated.

The average and maximum DGP-specific anchor-relative regrets were 0.0475 and 0.0668 for `top_bin_is_best`. Among the robust top-decile summaries, the corresponding pairs were 0.0508 and 0.0594 for the trimmed mean, 0.0525 and 0.0621 for the ordinary mean, 0.0592 and 0.0700 for the stability-penalized mean, and 0.0635 and 0.0852 for the median. All four top-decile summaries ranked second against the 23 individual estimators in each DGP.

Simulation truth entered only the post-selection evaluation. DGP-specific anchor benchmarks were `s_rf` for DGP1 and `x_rf` for DGP2. Regret relative to the corresponding benchmark and regret relative to the true-CATE oracle were summarized over replications. The two rank columns in Table S1 provide a separate descriptive comparison: the selected ensemble’s Monte Carlo mean true top-10% holdout CATE was ranked against the Monte Carlo means of all 23 individual estimators, yielding 24 displayed methods. Every displayed

individual estimator was also eligible for the ensemble path.

S1.6 Sensitivity to the first-stage ordering score

The 40 functions in Table S1 select one ensemble from a path already ordered by calibration slope. A separate sensitivity analysis changed the first-stage ordering score while holding the complete-support estimator set and second-stage rule fixed. For estimator m , define the finite-fold directional sign frequency

$$\bar{B}_{m,0.9} = \frac{1}{|\mathcal{F}_m|} \sum_{f \in \mathcal{F}_m} \mathbf{1}\{\Delta_{mf,0.10} > 0\}.$$

For each replication, the comparison retained only estimators with finite $\Delta_{mf,0.10}$, $t_{mf,0.10}$, and calibration slope on all 12 folds. The same complete-support set was used for both first-stage orderings. The $\bar{B}_{0.9}$ specification ranked that set from largest to smallest sign frequency, using $\bar{\Delta}_{0.10}$, the mean fold t -statistic, and estimator name as deterministic tie-breakers; the calibration-slope specification ranked the same set by estimator-level CalSlope_m . Each order generated nested equal-weight ensembles, and $\bar{\Delta}_{0.10}$ selected the path length. The eligible set contained 19–23 estimators in each DGP, with means of 21.9 in DGP1 and 22.8 in DGP2.

Under common support, the $\bar{B}_{0.9}$ -ordered ensemble has mean true top-10% holdout CATE of 0.0241 in DGP1 and 0.0682 in DGP2, compared with 0.0277 and 0.0708 under calibration-slope ordering. It ranks 11th of 24 displayed methods in DGP1 and 6th of 24 in DGP2, whereas the calibration-slope specification ranks second in both designs. These results describe the relative performance of the two first-stage scores in the simulation designs considered.

Table S2: Sensitivity to the First-Stage Ordering Score, with $\bar{\Delta}_{0.10}$ Fixed as the Second-Stage Score

First-stage score	DGP1		DGP2	
	Mean true CATE	Rank	Mean true CATE	Rank
Calibration slope	0.0277	2/24	0.0708	2/24
$\bar{B}_{0.9}$	0.0241	11/24	0.0682	6/24

Note: Mean true CATE is the Monte Carlo mean ground-truth CATE in the holdout subgroup above the transferred training-validation 90th-percentile threshold. Ranks compare each selected ensemble with the same 23 individual estimators. Within each replication, both specifications use the same estimators having complete support on all 12 folds and differ only in their first-stage ordering score.

The same complete-support rule was applied to the saved application fold-prediction libraries. The common eligible set contained 20 estimators for the bill-payment and debit-card arms and 17 for the availability-reminder comparison. The $\bar{B}_{0.9}$ ordering selected `r_rflasso` for the bill-payment arm, with a cached holdout ATE of 0.0585 (SE = 0.0341), and selected the same four-model ensemble as the primary procedure for the availability-reminder comparison, with a cached holdout ATE of -0.0112 (SE = 0.0306). The debit-card sensitivity path selected an ensemble containing causal-tree and causal-forest members. Their fold predictions were available for selection, but full-fit training-validation and holdout predictions were unavailable in the separate shared cache used for this post hoc calculation; consequently, no debit-card holdout estimate is reported for the alternative first-stage score. This cache limitation does not affect the simulation comparison or the primary calibration-slope application results.

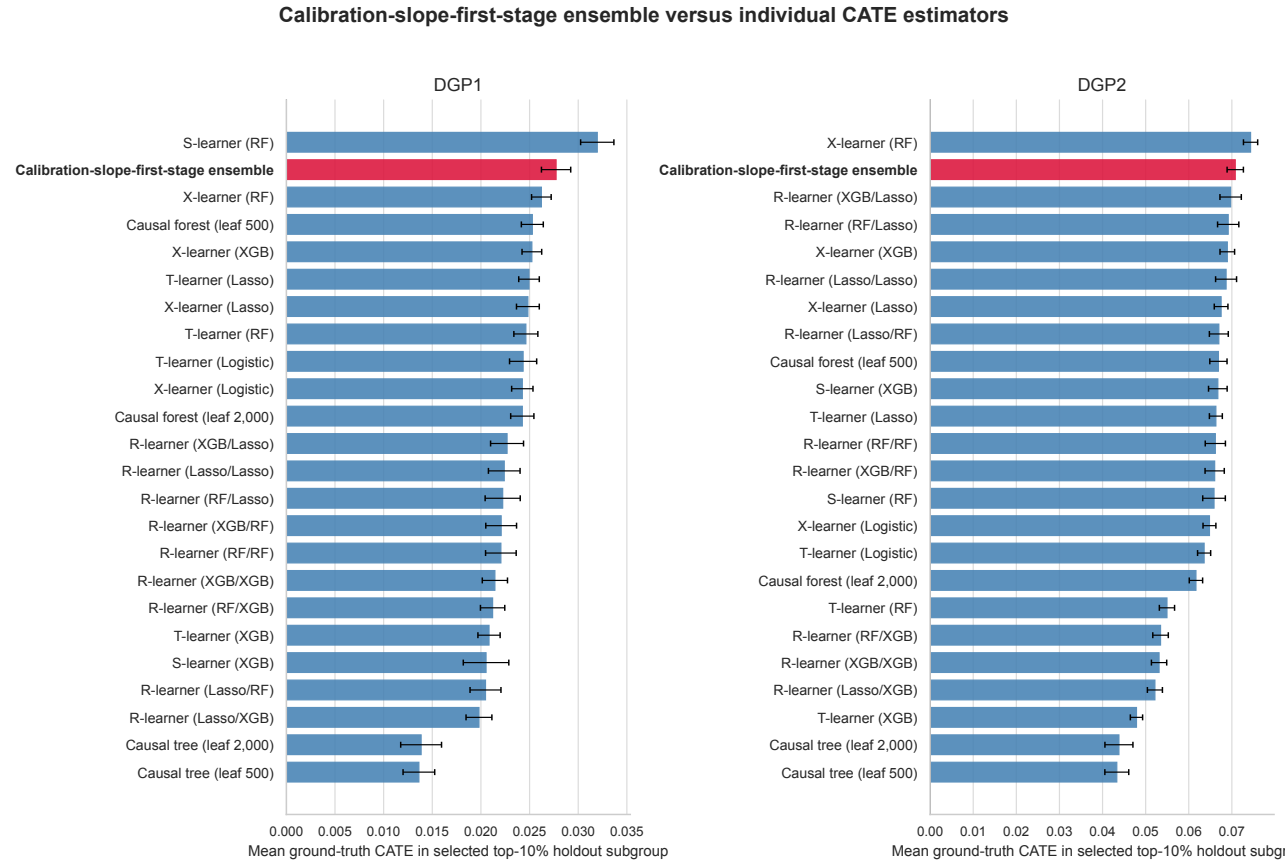


Figure S1: Calibration-slope-first-stage nested ensemble, restricted to estimators with complete fold support, compared with all 23 individual CATE estimators. Bars show mean ground-truth CATE above the transferred 90th-percentile holdout threshold; error bars equal 1.96 Monte Carlo standard errors of the mean. The highlighted ensemble is selected without true CATE and ranks second of 24 displayed methods by its Monte Carlo mean in both DGPs.

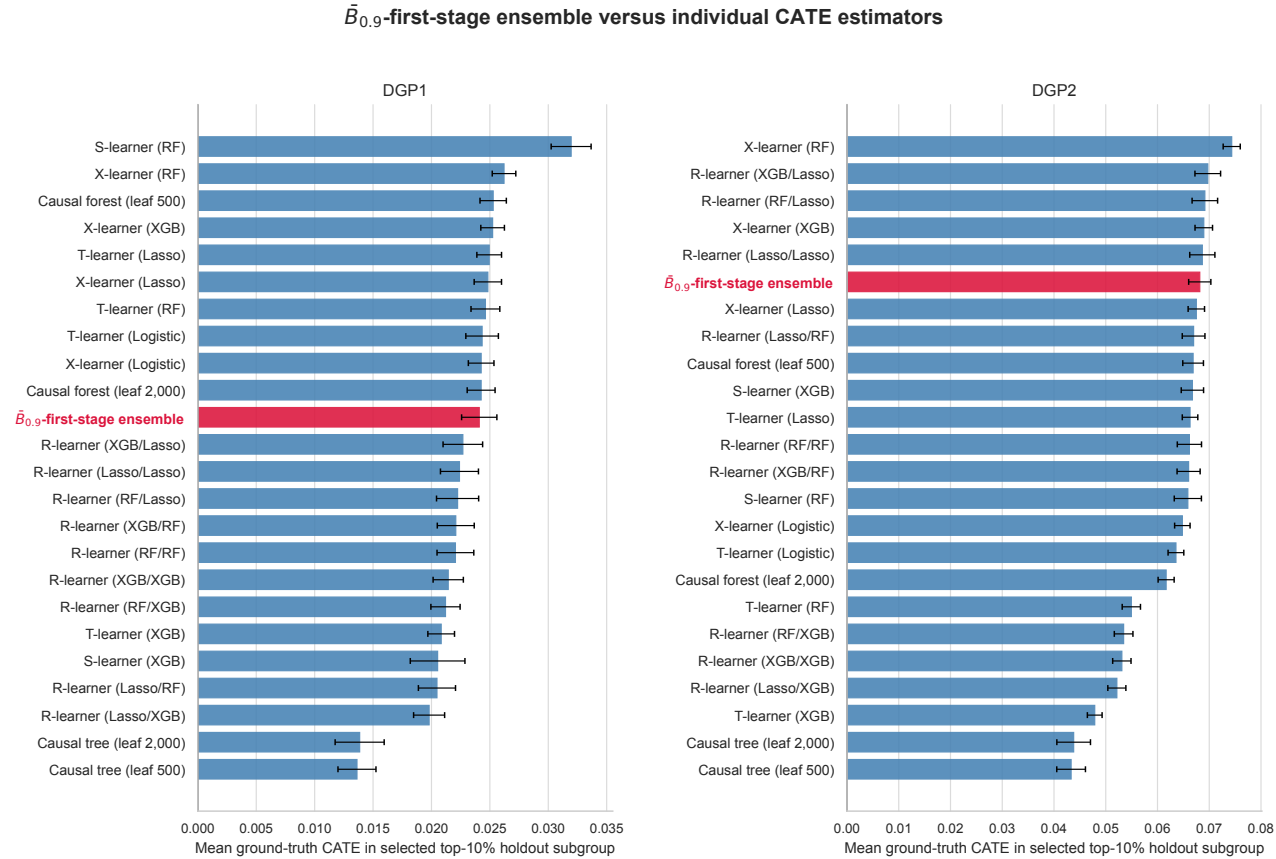


Figure S2: The $\bar{B}_{0.9}$ -first-stage nested ensemble, restricted to estimators with complete fold support, compared with all 23 individual CATE estimators. Bars show mean ground-truth CATE above the transferred 90th-percentile holdout threshold; error bars equal 1.96 Monte Carlo standard errors of the mean. The highlighted ensemble uses $\bar{B}_{0.9}$ only to order the nested path and uses $\bar{\Delta}_{0.10}$ to select its length. It ranks 11th of 24 displayed methods in DGP1 and 6th of 24 in DGP2.

S1.7 Exploratory shortlist and final statistic

The development sequence combined simulation screening with an exploratory holdout audit. That audit informed the choice of `mean_delta_top10` but was not an independent evaluation of the final set of 40 rules. The archived real-data result therefore reports only the fixed primary rule.

The initial shortlist contained `mean_delta_top10`, `neg_targeting_slope`, `qini`, `rate`, and `top_bin_is_best`. The three robust top-decile criteria were evaluated as simulation sensitivity checks and were not used to reopen the primary-rule choice. The trimmed mean has slightly lower simulation regret than the untrimmed mean, but changing the primary rule after observing this comparison would introduce another post hoc selection step.

The primary statistic remains `mean_delta_top10`. It directly matches the reported top-10% effect contrast and selected the two-model ensemble comprising `x_rf` and `r_rflasso` for the bill-payment arm. The holdout estimate and its 90% interval are descriptive of the selected subgroup in that split, but the nominal 10% threshold does not account for development-stage reuse of the holdout across statistics and treatment arms. The application result and simulation evaluations are therefore exploratory. Confirmatory evidence would require evaluating the complete rule on a new experiment or an untouched external sample.

S1.8 Reproducibility map

The development directory is `projects/causal-targeting-simulation`. The source output is `output/large_selection_bank_simfirst`, and the complete nonredundant bank is

written to `output/large_selection_bank_simfirst_40` by `build_40_stat_bank_from_cached_predictions.py`. The principal files are listed below.

- `sim_detail.csv`: replication-level results;
- `sim_summary.csv`: DGP-specific summaries;
- `simulation_winner_ranking.csv`: regret ranking; and
- `simulation_true_cate_ranks.csv`: ranks against individual estimators.

The complete 40-rule definition is in `screen_large_selection_bank_simfirst.py`.

A separate artifact-audit script checks the causal-tree and causal-forest fits. Its combined output contains 144 estimator-perturbation-fold records, all with finite fold predictions. Main-text Figure 5 is generated by `plot_primary_ensemble_true_cate_barchart.py` from the primary-procedure output and saved prediction caches.

The script `reproduce_bbar_first_stage_procedure.py` generates the first-stage sensitivity results from saved simulation and real-data fold predictions; it does not fit a CATE estimator. The associated $\bar{B}_{0.9}$ ranking figure is generated by `plot_bbar_first_stage_ensemble_true_cate_barchart.py`. Submission copies are included in the replication folder. The fixed application rule remains `mean_delta_top10` in `screen_anchor_augmented_procedure.py`.

S2 Other Treatment-Arm Comparisons

The same procedure was applied to two additional treatment-arm comparisons. For the debit-card bundled message against the baseline availability message, the selected ensemble

is `s_rf` and the holdout ATE is -0.0724 ($SE = 0.0362$; $t = -2.00$). For the availability reminder against no message, the selected ensemble is `{x_rf, x_lasso, t_lasso, x_logistic}` and the holdout ATE is -0.0119 ($SE = 0.0306$; $t = -0.39$). Neither comparison yields a positive holdout effect. As with the bill-payment comparison, these estimates are descriptive of the audited holdout splits.

Table S3: Final Validation Results for the Other Treatment Arms under the Calibration-Slope Nested-Ensemble Procedure

Treatment arm	Selected ensemble	Holdout ATE (SE), t
Debit-card bundled message	<code>s_rf</code>	-0.0724 (0.0362), -2.00
Availability reminder vs no message	<code>x_rf, x_lasso, t_lasso, x_logistic</code>	-0.0119 (0.0306), -0.39